

REFINEMENT OF THE AgI-Ag₂WO₄ PHASE DIAGRAM

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Silver-conducting solid electrolytes are promising materials for all-solid state batteries with a wide working temperature range, high stability, high safety, and low risk of leakage and corrosion, in contrast to batteries based on liquid organic electrolytes.

The system AgI-Ag₂WO₄ is known since early 1970s. There are two different versions of the phase diagram of the AgI – Ag₂WO₄ system [1, 2]. According to [1], it contains three chemical compounds: Ag₅IW₂O₈ (AgI/Ag₂WO₄=1:2), Ag₄I₂WO₄ (2:1) and Ag₆I₄WO₄ (4:1), while according to [2], there are only two – Ag₅IW₂O₈ and Ag₆I₄WO₄. All phases are ionic conductors, and the Ag₆I₄WO₄ compound exhibits superionic conductivity at ambient temperatures. Later, the composition of the superionic phase was represented by the formula Ag₂₆I₁₈W₄O₁₆ (4.5:1) [3]. Our repeated attempts to synthesize Ag₆I₄WO₄ and Ag₂₆I₁₈W₄O₁₆ compounds using the ampoule method (400°C, furnace cooling to 25°C) from AgI and Ag₂WO₄ each time resulted in non-single-phase products with a small amount of AgI impurity, which contradicts both versions of the phase diagram. Therefore, the goal of this study was to re-examine the phase equilibria in the AgI – Ag₂WO₄ system, refine its phase diagram, and establish the composition of the superionic phase.

Mixtures of the reagents AgI and Ag₂WO₄ in various molar ratios were prepared, sealed in ampoules and annealed according to the selected scheme. The phase composition of the products was analyzed using XRD and DSC. Single-phase products with the Ag₂₆I₁₈W₄O₁₆ structure, exhibiting superionic conductivity at room temperature, were obtained in a wide composition range from 3:1 to 3.9:1 (or 20.4...25.0 mol.% Ag₂WO₄); it corresponds to the general formula Ag_{2+x}I_xWO₄, where $x = 3 \div 3.9$. This region does not cover any of the superionic phase compositions described in the literature (Ag₆I₄WO₄ and Ag₂₆I₁₈W₄O₁₆) and is shifted toward a higher Ag₂WO₄ content. It is well known that solid electrolytes based on AgI, stabilized by oxoanions of the XO₄²⁻ type (where X is a transition metal) are prone to metastable states. A test performed by measuring the temperature dependence of conductivity in the thermal cycling mode confirmed the metastability of Ag_{2+x}I_xWO₄ except for 3.0:1. Based on the results obtained, it can be concluded that disagreements between the authors [1-3] are due to the fact that the metastable nature of phase equilibria was not taken into account. Changes and clarifications were made to the phase diagram of the AgI – Ag₂WO₄ system, and optimal conditions for the synthesis of single-phase products were selected.

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